

GULLFAKS C, Norway

WMO: 013000

Lat: 61.204N

Lon: 2.269E

Elev: 256

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.7	33.4	16.9	13.1	34.6	19.9	15.0	36.0	48.2	40.1	45.0	40.4	17.2	330	0.907

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	65.6	59.5	62.2	58.1	60.2	57.2	60.9	63.7	59.3	61.3	57.8	59.4	12.7	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.7	77.2	61.8	58.2	73.2	60.2	56.9	69.8	58.6	27.1	63.7	26.0	61.5	25.1	59.4	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 43.1	37.9	34.3	DB	28.3	70.2	2.4	4.2	26.6	73.2	25.2	75.7	23.8	78.0	22.1	81.1	(4)
(5)			WB	25.7	62.1	2.5	2.8	23.8	64.1	22.3	65.7	20.9	67.3	19.1	69.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	41.0	40.7	41.6	43.9	47.0	51.4	55.4	56.4	53.5	48.7	44.8	42.4
	DBStd	6.46	3.37	3.52	3.18	3.13	3.67	3.15	3.62	2.94	3.15	3.29	3.74	3.88
	HDD50	1565	278	260	262	183	107	16	1	0	7	61	156	234
	HDD65	6471	743	680	727	632	557	409	299	269	346	505	605	699
	CDD50	572	0	0	0	1	14	58	169	197	111	21	1	0
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	17.9	21.9	20.3	18.8	16.8	15.4	14.7	14.3	14.6	17.5	19.1	20.2	21.1
Precipitation	PrecAvg	52.8	6.3	5.2	5.0	2.8	1.9	2.6	3.2	3.9	4.6	5.6	5.5	6.1
	PrecMax	65.0	9.9	10.1	9.3	4.8	3.3	5.2	6.0	6.4	7.5	8.5	8.7	9.4
	PrecMin	37.6	3.0	2.3	1.8	1.0	0.9	1.0	1.6	2.6	1.9	2.3	1.7	2.6
	PrecStd	6.5	1.9	2.1	1.8	0.9	0.6	0.9	0.9	1.0	1.3	1.6	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	48.2	48.6	53.9	61.9	66.0	71.5	68.6	63.4	56.2	52.2	50.3
		MCWB	47.1	46.4	45.8	46.3	52.7	56.0	61.2	60.2	57.4	52.1	50.2	48.9
	2%	DB	47.6	47.0	47.3	50.5	57.0	60.8	67.3	63.9	60.3	54.8	51.2	49.3
		MCWB	46.1	45.2	45.1	45.7	50.0	54.6	60.8	59.8	56.1	51.9	49.3	47.9
	5%	DB	46.8	46.3	46.4	48.7	53.7	56.6	62.6	61.6	58.6	53.7	50.4	48.4
		MCWB	45.0	44.4	44.4	45.4	49.3	52.9	58.9	58.6	55.5	51.2	48.5	46.8
	10%	DB	46.0	45.5	45.6	47.6	51.4	54.7	59.9	60.0	57.4	52.8	49.6	47.6
		MCWB	43.9	43.4	43.4	44.9	48.4	51.8	57.8	57.3	55.3	50.4	47.6	45.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.0	47.5	47.5	48.3	54.6	59.1	63.5	62.5	58.6	54.6	51.7	49.8
		MCDB	48.2	47.7	47.8	49.9	60.1	63.4	68.9	64.9	61.0	54.9	52.0	50.0
	2%	WB	46.9	46.3	46.3	47.4	51.8	55.2	61.2	61.0	57.5	53.1	50.4	48.5
		MCDB	47.3	46.7	46.8	48.7	55.0	59.0	64.7	63.3	58.9	53.8	50.7	48.9
	5%	WB	45.7	45.2	45.1	46.6	50.1	53.8	59.7	59.3	56.7	52.1	49.3	47.5
		MCDB	46.6	46.0	45.8	47.8	52.3	55.6	62.4	60.7	57.9	53.0	50.0	48.2
	10%	WB	44.4	43.9	44.0	45.7	48.7	52.7	57.9	58.0	55.7	51.1	48.2	46.2
		MCDB	45.8	45.1	45.1	47.2	50.6	54.3	59.8	59.3	56.9	52.3	49.3	47.3
Mean Daily Temperature Range	5% DB	MDBR	4.8	4.5	4.4	4.5	4.5	4.0	4.0	4.0	3.9	3.9	4.4	4.7
		MCDBR	4.9	4.9	4.5	5.6	7.7	6.8	7.7	6.4	5.3	4.4	4.4	5.1
		MCWBR	6.2	5.6	5.1	4.0	4.7	4.3	4.2	4.1	3.8	4.6	5.4	6.5
	5% WB	MCDBR	4.8	4.6	4.3	4.6	6.4	5.2	6.5	5.3	4.4	4.0	4.6	5.1
		MCWBR	6.4	6.0	5.5	4.2	4.4	4.1	4.3	4.4	4.0	5.0	6.1	6.8
Clear-Sky Solar Irradiance	taub	0.325	0.349	0.363	0.377	0.365	0.358	0.375	0.367	0.361	0.341	0.313	0.382	
	taud	2.070	2.203	2.285	2.314	2.422	2.479	2.451	2.471	2.441	2.321	2.134	2.329	
	Ebn at Noon	112	192	237	258	272	276	267	260	239	195	117	70	
	Edh at Noon	14	22	29	33	32	31	31	28	24	19	12	9	
All-Sky Solar Radiation	RadAvg	59	217	580	1140	1595	1647	1484	1162	702	327	94	37	
	RadStd	8	31	78	112	108	184	120	99	57	44	13	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	+1.17	N/A	N/A	N/A	N/A	-149	-212	N/A	N/A

Nomenclature: See separate page